Pre-registered scientific criteria are specifications: eleven observed failures from one cosmology campaign, mapped onto known specification pathologies

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Abstract

We report eleven ways a hash-frozen success criterion failed during one intensive cosmology campaign — 89 frozen criteria, 201 register entries, 81 adjudicated attacks over two days — and we map every one of them onto a pathology already named outside our field. **None of the eleven is a new failure mode.** One of them — a validation passing at 10^{-4} relative while the quantity it converts does not exist — is *vacuity*, named by Beer et al. (1997) in 1997, whose canonical example ("every request is eventually acknowledged", satisfied by a system that never issues requests) is ours with a floating-point tolerance in place of a temporal operator. Another is code–comment inconsistency; another is a parameter on a physical boundary (Feldman & Cousins, 1998); another an unsatisfiable specification; another a flaky test on floating-point equality. Our claim is therefore a *mapping*, not a discovery: pre-registered scientific acceptance criteria are specifications, and they inherit the entire known pathology of specifications. We argue this is worth stating because the existing taxonomies of pre-registration failure (Yamada, 2018; Bakker et al., 2020) are *behavioural* — they classify what a researcher does — whereas the failures we observed are *structural*: they classify what the criterion *is*, and every one occurred with no misconduct whatever. We give the instance count, the class each instance belongs to, and which existing tooling would have caught it. We report one protocol invariant for which we could find no prior art — the criteria block is immutable while the script body stays freely mutable — and we explicitly decline to claim the rest of the protocol, which is an integration of OSF-style registration, continuous-analysis CI (Beaulieu-Jones & Greene, 2017) and design by contract. Finally we report the one failure that is not a criterion failure at all: after three frozen designs blocked on the same question, a fourth would have passed, and writing it would have been p-hacking one level up. Hash-freezing protects each study; it does not protect the series.

1 The protocol, reported as a design note

We state the mechanism briefly, and we do not claim it. The unit is a script. Its docstring opens with a block of success criteria: the validations that must pass before anything is published, the criteria themselves — required *exhaustive* and *exclusive*, so that every outcome maps to exactly one written verdict — and a declaration of what the study will not claim. `freeze` extracts the docstring by AST and records its SHA-256; `verify` fails CI if the hash moved; amendments are permitted but written to a public log with a mandatory justification.

Every component is off the shelf. Criteria inside the executable artefact is design by contract, and in Python specifically it is `doctest`. Hash-freezing an analysis plan is OSF registration, and in cryptographic form OriginStamp (Gipp et al., 2018) and CryptSubmit (Gipp et al., 2017); a search of OpenAlex returns of order a hundred physics and cosmology deposits already doing it, several from 2026 combining a SHA-256 manifest with an OpenTimestamps attestation. CI

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enforcement of a research artefact is continuous analysis (Beaulieu-Jones & Greene, 2017) and the Popper convention (Jimenez et al., 2017). A public amendment log with justification is OSF's withdrawal mechanism, and reporting deviations has its own guidance literature (Willroth & Atherton, 2024). Fixing acceptance criteria before looking is blind analysis, standard in particle physics for two decades (Klein & Roodman, 2005; MacCoun & Perlmutter, 2015) and in cosmology since DES (Muir et al., 2020).

The one invariant we could not find elsewhere. Prior art freezes the *artefact* — the plan, the code, the manifest. We freeze only the *acceptance predicate*, and leave the body freely mutable. That asymmetry is what makes correction and pre-registration compatible: a script may be debugged, re-parameterised and re-run without touching what would count as success. Over 89 frozen criteria the amendment channel was used zero times — not because the criteria were always right, but because being wrong was survivable without editing them. We offer this asymmetry as the protocol's only contribution and would not defend more.

2 The thesis: criteria are specifications

A pre-registered criterion is a predicate a computation must satisfy for a claim to count as established. That is what a specification is. The consequence is immediate and, as far as we can find, unstated: **pre-registered scientific criteria inherit the entire known pathology of specifications** — vacuous satisfaction, unsatisfiability, drift from the implementation, insufficient sensitivity — together with the detection literature attending each.

We arrived at this the slow way. We wrote eleven criteria that failed, catalogued them as though they were new, and discovered on checking that every one already had a name.

3 Eleven observed instances, and the class each belongs to

Table 1 is the paper. Three observations follow.

Vacuity is the one that should worry practitioners most. Beer et al. (1997) define vacuity as a specification "satisfied too easily, or vacuously", and warn that it "misleads users of model-checking into thinking that a system is correct". Their canonical example — "every request is eventually followed by an acknowledgment" is satisfied vacuously by a system that never generates requests — is precisely our instance, with a tolerance in place of a temporal operator. Our validation passed at 10^{-4} relative and established nothing. *A criterion that passes perfectly while measuring a phantom is more dangerous than one that fails*, and formal methods has known this since before the pre-registration movement existed. To our knowledge this is the first reported instance of vacuity in a pre-registered *scientific* acceptance criterion; that is a far smaller claim than a new failure mode, and it is the true one.

Existing tooling would have caught four of the seven classes. Specification drift is detectable by code–comment inconsistency tools; unsatisfiability by a satisfiability check on the criteria block; floating-point flakiness by a lint; insufficient sensitivity by the order-of-accuracy discipline of code verification. Vacuity has dedicated algorithms. None of this tooling is applied to pre-registered scientific criteria today, and we know of no reason it could not be.

What no tooling would have caught. Non-identifiability, the boundary minimum and the weak comparator are statistical, not syntactic. The boundary minimum in particular was found only because a human reader remarked that a result looked *too good to be true*; every automated check had passed it.

4 How this differs from existing pre-registration taxonomies

Yamada (2018) gives four ways to "crack" pre-registration: selective reporting, infinite re-experimenting, overissuing, and pre-registering after results are known. Bakker et al. (2020) enumerate 29 researcher degrees of freedom and find that structured formats reduce but do not eliminate them; Claesen et al. (2021) audit adherence in the first generation of pre-registered studies; Wicherts et al. (2016) give a 34-item checklist.

These are *behavioural* taxonomies: they classify what a researcher does, and their modes are, in Yamada's own framing, things a malicious researcher might do. The failures we report are *structural*. They classify what the criterion *is*, and every one occurred with no misconduct: each was written by an analyst trying to be rigorous, each passed a reading at the moment it was frozen, and several were caught by the analyst's own machinery days later. That distinction is the gap we claim, and it is the only one.

One adjacency is not a coincidence. Yamada's "infinite re-experimenting" — repeating an experiment until the data cooperate — is the behavioural sibling of §5, where designs rather than datasets are repeated until one passes. The family resemblance is exact; the object differs.

5 The failure that is not a criterion failure

One question in the campaign was attempted three times. The first design blocked on a grid floor. The second, with a ten-fold finer grid, blocked because the floor had been *displaced* rather than lifted. The third replaced the grid-walk estimator with a self-validating parabolic fit — which worked, with no floor at all, its uncertainty scaling as $\times 2.00$ then $\times 4.00$ exactly — and blocked because its own validation demanded exact reproduction of the *grid* values the new estimator had deliberately abandoned. That last block belongs to the unsatisfiability class, and we authored it after cataloguing that class.

A fourth design tolerating one grid step would have passed; every other validation already did, and the answer it would have returned was visible in the third design's output. We did not write it.

Hash-freezing protects each study in isolation and does not protect the series. What is frozen is every attempt, not the decision to attempt again. Iterating designs until one passes is p-hacking one level up, and pre-registration as usually practised has no defence against it — each attempt is impeccably pre-registered. The only guard we can see is a declared stopping rule. We adopted one, *at the third block on the same question, stop and record the status as not established*; we note that it is arbitrary, that its whole value lies in being public, and that we adopted it only after needing it. The cost is real: a signature that stayed strictly invariant under a four-fold reduction in uncertainty remains unusable. We prefer that cost to a verdict obtained by design selection.

6 Practices we adopted, and their prior art

We list these for completeness and attribute each; none is ours.

Mutation-test the verifier before believing its pass rate. A tool returning "55 of 55, no blocking failure" establishes nothing until it is shown capable of failing; we injected five faults and required all five to be caught. This is mutation testing (Jia & Harman, 2010) applied to the oracle rather than the code, which is Knauth et al. (2009); and mutating checkers specifically to detect vacuity is Vacuity by checker mutation (2010) — that is, the combination of our vacuity class with this practice was published in 2010. It has also been applied to scientific software (Hook & Kelly, 2009). We report only that we found it necessary.

Detect generically, never from a hand-written list. A template check enumerating five placeholder names by hand passed while a sixth, added later, went unsubstituted and broke the published page at its first line. The replacement is one regular expression and cannot miss a placeholder it has never heard of.

Test the refuting branch before the confirming one, and set the confirmation threshold stricter than the refutation threshold. Twice this ordering mattered; once it returned a refutation of a mechanism we had asserted in two previous register entries without ever testing it.

Declare the direction of every substitution. Where a value must be assumed, choose it against the thesis being defended, and say that this is why.

7 Limits

$N = 1$: one corpus, one analyst, two days. We cannot say how the eleven instances distribute in general, only that they occurred and that none required bad faith. The instance count is the empirical content: 89 criteria frozen, eleven failures of the kinds above, and five studies that halted on their own validations and published nothing.

The protocol is also mechanical. It does not read equations and does not judge physics. In this campaign the most consequential single audit was triggered by a human reader, and a literature verification killed a planned companion paper outright by establishing that all four of its claimed results were already published — the oldest in 2015, the newest four days earlier. Neither is reachable by hashing. The same kind of verification produced this paper's final form: it established that none of our eleven failures was a new mode, and the paper is better for it.

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observed instance	named class	where the name comes from
Criterion frozen on a quantity the data cannot constrain; satisfiable by any outcome.	**Non-identifiability**	standard estimation theory
A goodness-of-fit bound $\chi^2/(N+17)$ frozen for a 22-bin dataset; the offset came from a different data configuration.	**Reduced-χ^2 misuse**	Andrae et al. (2010)
A validation comparing the model at a *fixed* parameter point against a reference at its *optimum*; the gap it measured was the optimisation. A validation run against a comparator that reintroduced a term the model had removed.	**Weak or mismatched comparator** — these two are one class, and a referee would say so	Ferrari Dacrema et al. (2019); the comparator-bias domain of clinical risk-of-bias instruments
A tight constraint whose minimum sat on the edge of the accessible domain; one branch was closed by the implementation, so a one-sided result was reported as two-sided.	**Parameter on a physical boundary**	Feldman & Cousins (1998), whose entire subject is one- versus two-sided intervals at a boundary
Two requirements frozen in one block — a 41-point grid and a ± 0.3 tolerance — that could not both be met. The study could not succeed by construction.	**Unsatisfiable specification**	Rozier & Vardi (2007); satisfiability checking is the standard sanity check
A docstring asserting "a dense grid (60 000 points, log)" while the code, after successive body corrections, used 240 001. A verification tool frozen with a sky-region definition differing by 10° in right ascension from the one the corpus used everywhere else.	**Specification drift** / code–comment inconsistency	Panthaplackel et al. (2021), and an active detection subfield
A validation requiring a conversion factor constant to 2%; it passed at 0.010%. The quantity it converted did not exist.	**Vacuity**	Beer et al. (1997); Kupferman & Vardi (2003); twenty-five years of detection algorithms
Two thresholds compared with `>=` on floating-point grid arithmetic; true values exactly 2 and 1, computed values 1.9999999999999791 and 0.9999999999999836. Both failed, both in the direction convenient to the analyst.	**Flaky test**, floating-point equality	Luo et al. (2014); a standard lint
A validation requiring an uncertainty to shrink by a factor k; the scan grid could not resolve an uncertainty below one step, so the requirement receded as fast as the quantity it measured.	**Insufficient test sensitivity**	code verification uses order-of-accuracy under grid refinement, not a single-grid tolerance (Salari & Knupp, 2000; Roache, 1998)

Table 1: Eleven instances observed in one campaign, mapped onto seven named classes. The compression from eleven to seven is deliberate and we perform it ourselves: presenting them as